Parametrised Recursion in Dialectica Categories

DRAFT

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Abstract

The Dialectica interpretation runs on Gödel's System T, so one could expect the dialectica categories to carry an object of natural numbers. Since they are not cartesian closed, the notion to ask for is the parametrised one: without a parameter the counterexample coordinate of the mediating morphism has nothing to carry, and only the witnesses are tested. Over a cartesian closed category $\mathcal{C}$ with a natural numbers object $\mathbb{N}$, and with relations taken as subobjects, the object $(\mathbb{N}, 0, 0)$ is a parametrised natural numbers object for the cartesian structure of Dial($\mathcal{C}$) and $(\mathbb{N}, 1, \top)$ is one for its tensor product, the witness coordinate of the mediating morphism iterating while the counterexample coordinate recurs backwards along the witnesses that iteration produces. In the Girard variant of the Dialectica, Gir($\mathcal{C}$), the cross tensor has such an object, its counterexample coordinate being the final coalgebra of streams, while over sets the products have none. In sets the parametrising objects are countable copowers, and over $\mathcal{C}$ the same formulas need no infinite coproducts. Nothing classical is used: what the verification needs from $\mathcal{C}$ is that its subobjects carry intuitionistic first-order logic. Any natural numbers object in the sense of Paré and Román is a cocommutative comonoid, whatever the ambient monoidal category, so the objects found for the tensors are duplicable and discardable, and each is a retract of its own !-version, so none of them is linear. In a topos a natural numbers object is the initial algebra of $1 + (-)$, and over sets both categories do have the initial algebras for the corresponding functors $T + (-)$ and $I + (-)$, given by those same two objects, together with the final coalgebra $(1, \mathbb{N}, \emptyset)$ for $J \times (-)$, which in the Girard variant is the de Morgan dual of the second. Initiality does not imply the parametrised property here, since $(\mathbb{N}, \emptyset, \emptyset)$ is an initial algebra for $T + (-)$ in Gir(**Sets**) and is not a parametrised natural numbers object there.

1 Introduction

Lawvere's elementary theory of the category of sets [10] axiomatises sets without membership, replacing each set-theoretic axiom by a universal property. The axiom of infinity becomes the existence of an object N together with a point $z\colon 1 \to N$, playing the part of zero, and an endomap $s\colon N \to N$, playing the part of the successor function, such that every other object carrying such a pair receives a unique map from this one. The universal property is a principle of definition by recursion rather than a list of assertions about N: a starting value $b\colon 1 \to B$ and a step $g\colon B \to B$ determine a map $N \to B$, and determine it uniquely.

Peano's axioms sit alongside this rather than inside it. Freyd showed that in a topos the two packages meet, a triple $(\mathbb{N}, 0, s)$ being a natural numbers object exactly when $[0, s]\colon 1 + \mathbb{N} \to \mathbb{N}$ is an isomorphism and $\mathbb{N} \rightrightarrows \mathbb{N} \to 1$ is a coequalizer of the identity and the successor; the first condition says that each element is zero or a successor in exactly

one way, and the second does the work of induction. A proof is in Johnstone [9, D5.1]. The dialectica categories are not toposes and there is no such equivalence to appeal to here, so throughout this note a natural numbers object is an object with the recursion property and nothing more, and induction is a separate matter, taken up in Section 7.

Burroni [2] observes that Lawvere's way of putting the matter asks very little of the ambient category, a terminal object and nothing more, while the force of the axiom in a topos comes from the surrounding structure. He proposes instead an axiom in which the source of z is an arbitrary object, so that the recursion carries a parameter, and calls a category satisfying it a Peano–Lawvere category. Paré and Román [13] take the complementary step of replacing the cartesian structure by a monoidal one, with the unit of the tensor in the role of the terminal object, their method being to pass to the cocommutative comonoids, where the cartesian theory is available. Their interest was in monoidal categories at large rather than in linear logic, and the definition reaches the linear logic literature with Mackie, Román and Abramsky [12], who use it in an internal language for autonomous categories.

The question here is what recursion looks like in the dialectica categories of [6], where an object is a predicate presented as a relation between witnesses and counterexamples, and a morphism carries witnesses forwards and counterexamples backwards. The Dialectica interpretation is an interpretation of arithmetic and runs on System T, so recursion is already present in what these categories were built to model, and one could expect Dial(**Sets**) to carry an object of natural numbers with something of the interpretation's own recursion in it. Section 5 shows that it does. The witness coordinate iterates as it does in sets, and the counterexample coordinate carries a second recursion that travels back down the sequence of witnesses the first one produced, which is available because the backward map of a dialectica morphism is allowed to see the witness it is refuting. That second recursion is not merely like the one the interpretation performs: it is, up to a single application of the backward map of the base morphism, the functional by which Gödel's interpretation handles induction, and Section 7 sets the two side by side.

Seeing this requires the parameter. Our earlier note [7] asks Lawvere's question, whose zero comes from the unit, and finds $(\mathbb{N}, \emptyset, \emptyset)$ for the cartesian structure and $(\mathbb{N}, 1, \top)$ for a tensor. With no parameter the objects one can map out of are restricted before the question is asked: from the terminal object only those with no counterexamples at all, and from the tensor unit only those with a witness that meets every counterexample, which is Fact 1 of [7]. The backward map of a mediating morphism then has empty or one-point data to produce, and the second coordinate can carry nothing. Burroni's parameter restores it, and since neither Dial(**Sets**) nor Gir(**Sets**) is cartesian closed the parameter cannot be transposed away, so the question has to be asked separately for the products and for each tensor. Asked that way of Gir(**Sets**), where we also looked, the answers come out differently: no parametrised object for the products, and for the cross tensor an object whose counterexample coordinate holds a sequence and coiterates. The contrast with Dial(**Sets**) is one of expectation rather than of capability, since there the recursion was something to look for, Gödel's interpretation needing it, while here nothing said in advance what would be found.

The computations do not need sets. Charles Morgan noticed, in an unfinished 2013 note, that a parametrised natural numbers object in the Girard variant over sets follows from Burroni's theorem that a category with countable copowers is a Peano–Lawvere category, and that its counterexample maps are the head and tail of streams. Read this way, each of the three positive answers is a copower: in Dial(**Sets**) the countable

copower of A has the witnesses $U \times \mathbb{N}$ and the counterexamples of A unchanged, and in Gir(**Sets**) it has the counterexamples $X^{\mathbb{N}}$, a stream object. Over a cartesian closed category $\mathcal{C}$ with a natural numbers object neither needs countable coproducts, since $U \times \mathbb{N}$ and $X^{\mathbb{N}}$ are already objects of $\mathcal{C}$, and the recursions that define the mediating morphisms are primitive recursions in exponentials of $\mathcal{C}$. We therefore work over such a $\mathcal{C}$, with relations as subobjects, and the verification that the mediating morphisms are morphisms of dialectica objects becomes an induction in the internal logic of $\mathcal{C}$. That induction is intuitionistic, so the results hold in any topos, and what we have not done is the corresponding calculation when the relations take values in a lineale.

Section 8 then records what the Paré–Román definition commits one to in any monoidal category, namely a comonoid; the observation is from an unpublished manuscript of Maria Emilia Maietti, Valeria de Paiva and Eike Ritter [5], where it is made syntactically, and the categorical proof given here needs only the uniqueness clause with a trivial parameter. The note follows the question in that order. Section 4 sets out what we need from $\mathcal{C}$, Sections 5 and 6 find the objects for each structure and each shape of morphism, Section 7 explains the recursion they carry by setting it beside Gödel's functional for induction, Section 8 states what the monoidal definition limits that recursion to, and Section 9 compares the parametrised property with initial algebras and final coalgebras.

2 Dialectica categories

There are two constructions in [6], differing in the shape of the backward map of a morphism, and both belong to the Dialectica family. We recall them over **Sets**, where they are easiest to read, and Section 4 takes them over a cartesian closed category.

Definition 1. Objects of Dial(**Sets**) and of Gir(**Sets**) are triples $A = (U, X, \alpha)$ with U and X sets and $\alpha \subseteq U \times X$ a relation. A morphism $A \to B = (V, Y, \beta)$ in Dial(**Sets**) is a pair of functions $f\colon U \to V$ and $F\colon U \times Y \to X$ such that

$$u\,\alpha\,F(u,y) \implies f(u)\,\beta\,y \qquad \text{for all } u \in U,\ y \in Y.$$

A morphism $A \to B$ in Gir(**Sets**) is a pair $f\colon U \to V$ and $F\colon Y \to X$ satisfying $u\,\alpha\,F(y) \Rightarrow f(u)\,\beta\,y$.

We read U as a set of witnesses, X as a set of counterexamples, and α as a relation of satisfaction between them. A morphism maps witnesses forwards and counterexamples backwards, and the condition says that the backward map turns a counterexample to the conclusion into a counterexample to the premiss.

Both categories have finite products. For $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$,

$$A \times B = (U \times V,\ X + Y,\ \mathrm{ch}),$$

where $(u, v)\,\mathrm{ch}\,(x, 0)$ iff $u\,\alpha\,x$ and $(u, v)\,\mathrm{ch}\,(y, 1)$ iff $v\,\beta\,y$. The terminal object is $(1, \emptyset, \emptyset)$.

The tensor of Dial(**Sets**) is

$$A \otimes B = (U \times V,\ X \times Y,\ \alpha \otimes \beta), \qquad (u, v)\,\alpha \otimes \beta\,(x, y) \ \text{ iff } \ u\,\alpha\,x \text{ and } v\,\beta\,y,$$

and the tensor of Gir(**Sets**) is

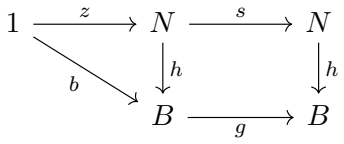
$$A \otimes_{\mathrm{Gir}} B = (U \times V,\ X^V \times Y^U,\ \alpha \otimes_{\mathrm{Gir}} \beta), \qquad (u, v)\alpha \otimes_{\mathrm{Gir}} \beta(\varphi, \psi) \ \text{ iff } \ u\alpha\varphi(v) \text{ and } v\beta\psi(u).$$

Each is symmetric monoidal closed with unit $I = (1, 1, =)$, the internal hom being right adjoint to the tensor in each case.

3 Parametrised natural numbers objects

Lawvere's definition asks for maps out of N itself, and we state it first.

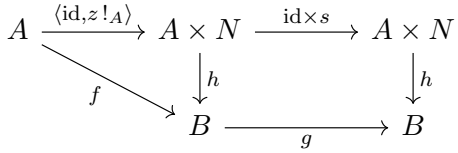
Definition 2 (Lawvere)**.** Let $\mathcal{E}$ have a terminal object. A *natural numbers object* is an object N with maps $z\colon 1 \to N$ and $s\colon N \to N$ such that for every object B and all $b\colon 1 \to B$, $g\colon B \to B$ there is a unique $h\colon N \to B$ making



commute.

A map out of N in a dialectica category is a pair of functions, and when zero comes from the unit the backward one has only the counterexamples of the codomain to work with. Burroni's formulation gives that backward map a second argument, and in Dial(**Sets**) the second argument is what the recursion in the counterexample coordinate needs. His axiom asks for a left adjoint to the forgetful functor from objects with an endomorphism, which produces a free object N_X over each X rather than a product; in the presence of finite products the two agree, and the formulation we use below, with $A \times N$, is the one standard in the literature on categories that are not cartesian closed, as in Johnstone [9, A2.5.2]. The monoidal version replaces the product by the tensor and the terminal object by the unit, as in Paré and Román.

Definition 3. Let $\mathcal{E}$ have finite products. A *parametrised natural numbers object* is an object N together with $z\colon 1 \to N$ and $s\colon N \to N$ such that for all objects A, B and all morphisms $f\colon A \to B$ and $g\colon B \to B$ there is a unique $h\colon A \times N \to B$ making



commute. If $\mathcal{E}$ is monoidal with unit I, a *parametrised natural numbers object for the tensor* is an object N with $z\colon I \to N$ and $s\colon N \to N$ such that for all such f and g there is a unique $h\colon A \otimes N \to B$ with $h \circ (\mathrm{id}_A \otimes z) \circ \rho_A^{-1} = f$ and $h \circ (\mathrm{id}_A \otimes s) = g \circ h$, where $\rho_A\colon A \otimes I \to A$ is the right unitor.

Taking A to be the terminal object, respectively the unit, recovers Lawvere's condition, and in a cartesian closed category the two are equivalent by transposing h across the adjunction. Neither Dial(**Sets**) nor Gir(**Sets**) is cartesian closed, so for them the parametrised form is the stronger demand.

4 The base category

Over sets the recursions below can be checked pointwise. We now ask what they need from a general base, so that the results hold for the dialectica categories built over any cartesian closed category with a natural numbers object, as the construction of [6] allows. Throughout, $\mathcal{C}$ is a cartesian closed category with finite limits and a natural numbers

object $(\mathbb{N}, 0, s)$. In a cartesian closed category a natural numbers object is automatically parametrised, so iteration with parameters is available, and with it primitive recursion. An object of $\mathrm{Dial}(\mathcal{C})$ or $\mathrm{Gir}(\mathcal{C})$ is a triple $A = (U, X, \alpha)$ with α a subobject of $U \times X$. A morphism $(f, F)\colon A \to B = (V, Y, \beta)$ of $\mathrm{Dial}(\mathcal{C})$ has $f\colon U \to V$ and $F\colon U \times Y \to X$ with

$$\langle \pi_U, F\rangle^* \alpha \;\leq\; (f \times Y)^* \beta \qquad \text{in } \mathrm{Sub}(U \times Y),$$

and a morphism of $\mathrm{Gir}(\mathcal{C})$ has $F\colon Y \to X$ and $(U \times F)^*\alpha \leq (f \times Y)^*\beta$. The tensors and the monoidal unit are given by the formulas of the previous sections, the exponential X^V now being the one of $\mathcal{C}$. For the products we also ask $\mathcal{C}$ for finite coproducts that are disjoint, so that the relation ch on $U \times V \times (X + Y)$ can be defined summand by summand; they are stable under pullback because $\mathcal{C}$ is cartesian closed, and the initial object 0 is strict for the same reason. The terminal object of $\mathrm{Dial}(\mathcal{C})$ and of $\mathrm{Gir}(\mathcal{C})$ is then $T = (1, 0, 0)$.

We further assume that each $\mathrm{Sub}(P)$ is a Heyting algebra and that pulling back along a projection $P \times Y \to P$ has a right adjoint $\forall_Y$, all of this stable under pullback, as in any topos. These are the assumptions under which the first-order intuitionistic logic of $\mathcal{C}$ is sound, and the proofs below are written in element notation and read in that logic: a statement such as $u\,\alpha\,F(u,y) \Rightarrow f(u)\,\beta\,y$ for all u and y is the inequality of subobjects displayed above. Nothing classical is used. The existence and uniqueness of the mediating morphisms need only the closed structure and the natural numbers object. The Heyting structure is used only to check that the morphisms so defined satisfy the dialectica condition, by an induction whose statement quantifies over all counterexamples, since the counterexample changes at every step.

Lemma 1. *Let P be a subobject of $U \times \mathbb{N}$ that contains the image of $\langle \mathrm{id}, 0\,!_U\rangle\colon U \to U \times \mathbb{N}$ and is closed under $\mathrm{id} \times s$. Then P is all of $U \times \mathbb{N}$.*

Proof. Write $m\colon P \to U \times \mathbb{N}$ for the inclusion, $p\colon U \to P$ for the factorisation of $\langle \mathrm{id}, 0\,!_U\rangle$ through m, and $t\colon P \to P$ for the restriction of $\mathrm{id} \times s$. Iteration with parameter U gives $k\colon U \times \mathbb{N} \to P$ with $k\langle \mathrm{id}, 0\,!_U\rangle = p$ and $k(\mathrm{id} \times s) = tk$. Then mk and the identity of $U \times \mathbb{N}$ both satisfy the equations for the base $\langle \mathrm{id}, 0\,!_U\rangle$ and the step $\mathrm{id} \times s$, so they are equal by uniqueness, and the monomorphism m is split epi, hence an isomorphism. □

Lemma 2. *Let $h\colon U \times \mathbb{N} \to V$, $F\colon U \times Y \to X$ and $G\colon V \times Y \to Y$ be morphisms of $\mathcal{C}$. There is a unique $H\colon U \times \mathbb{N} \times Y \to X$ with*

$$H(u, 0, y) = F(u, y), \qquad H(u, n+1, y) = H\big(u,\, n,\, G(h(u, n),\, y)\big).$$

Proof. Transposing, H is a morphism $\widehat{H}\colon U \times \mathbb{N} \to X^Y$ with $\widehat{H}(u, 0) = \lambda y.\, F(u, y)$ and $\widehat{H}(u, n+1) = \lambda y.\, \widehat{H}(u, n)\big(G(h(u, n), y)\big)$. The step uses u and n as well as the previous value, so we iterate with parameter U on $U \times \mathbb{N} \times X^Y$, with base $u \mapsto (u, 0, \lambda y.\, F(u, y))$ and step $(u, n, \varphi) \mapsto (u,\, n+1,\, \lambda y.\, \varphi(G(h(u, n), y)))$. By uniqueness the first two components of the resulting morphism are the projections, and its third component is $\widehat{H}$. Any H' satisfying the two equations gives, by the same pairing, another morphism satisfying the equations of the iteration, so $H' = H$. □

All the calculations in this note take relations to be subobjects of $\mathcal{C}$, so the dialectica condition is checked by induction in $\mathrm{Sub}(-)$. When the relations take values in a lineale instead, in the sense of Martin Hyland and Valeria de Paiva [4], the condition has to be checked in the lineale itself, and whether the corresponding induction holds there depends on the lineale. We have not yet done that calculation.

5 Products

The cartesian structure is where Lawvere asked the question and where the earlier note found its first answer, so it is the place to start. Both categories have the same products and the same terminal object $T = (1, 0, 0)$, and in both a morphism out of T has a backward component landing in the initial object, so the candidate object comes out the same in each, and the backward maps available to factor through it do not.

Proposition 3. *The object $N = (\mathbb{N}, 0, 0)$, with $z = (0, !)$ and $s = (s, !)$, is a parametrised natural numbers object for the cartesian structure of* $\mathrm{Dial}(\mathcal{C})$.

Proof. A morphism $(1, 0, 0) \to (W, M, \mu)$ has a backward component $1 \times M \to 0$, so M is initial, 0 being strict, and with it μ. Any candidate is therefore of the form $(W, 0, 0)$, and since morphisms between objects of this form are just morphisms of $\mathcal{C}$ between their witnesses, W is forced to be a natural numbers object of $\mathcal{C}$.

Let $A = (U, X, \alpha)$. Since the second component of N is initial, $A \times N \cong (U \times \mathbb{N},\ X,\ \alpha')$ with $(u, n)\ \alpha'\ x$ iff $u\ \alpha\ x$, that is, α pulled back along the projection. Let $B = (V, Y, \beta)$ and let $(f, F)\colon A \to B$ and $(g, G)\colon B \to B$, so that $F\colon U \times Y \to X$ and $G\colon V \times Y \to Y$. A morphism $A \times N \to B$ is a pair $h\colon U \times \mathbb{N} \to V$ and $H\colon U \times \mathbb{N} \times Y \to X$ subject to

$$u\ \alpha\ H(u, n, y) \implies h(u, n)\ \beta\ y. \tag{1}$$

Read coordinatewise, the triangle says

$$h(u, 0) = f(u), \qquad H(u, 0, y) = F(u, y),$$

and the square says

$$h(u, n+1) = g\big(h(u, n)\big), \qquad H(u, n+1, y) = H\big(u,\ n,\ G(h(u, n),\ y)\big).$$

Both components are thereby determined, h by parametrised iteration in $\mathcal{C}$ and H by Lemma 2, so a mediating morphism is unique if it exists. It remains to check (1) for all y, that is, that the subobject of $U \times \mathbb{N}$ given by $\forall y\,(u\ \alpha\ H(u, n, y) \Rightarrow h(u, n)\ \beta\ y)$ is everything. By Lemma 1 this comes to the two cases below.

For $n = 0$, suppose $u\ \alpha\ H(u, 0, y) = F(u, y)$. Then $f(u)\ \beta\ y$ because (f, F) is a morphism, and $f(u) = h(u, 0)$.

For $n+1$, suppose $u\,\alpha\,H(u, n+1, y) = H\big(u, n, G(h(u, n), y)\big)$. The induction hypothesis, applied to the counterexample $G(h(u, n), y)$, gives $h(u, n)\ \beta\ G(h(u, n), y)$. Putting $v = h(u, n)$ in the condition on (g, G), namely $v\,\beta\,G(v, y) \Rightarrow g(v)\,\beta\,y$, gives $g(h(u, n))\,\beta\,y$, which is $h(u, n+1)\ \beta\ y$. □

The induction step is where the shape of a dialectica morphism does its work. The backward map G is applied at the witness $h(u, n)$ that the computation has reached after n steps, and the dependence of the backward map on the forward one is what makes the induction hypothesis available in the form the step needs.

A counterexample to the value at n runs the computation backwards. Unfolding the recursion for H,

$$H(u, n, y) \;=\; F\Big(u,\ G\big(h(u, 0),\ G(h(u, 1),\ \dots\ G(h(u, n-1),\ y)\dots)\big)\Big),$$

so a counterexample to the value at n is transported down through the trace $h(u, n-1), \dots, h(u, 0)$ and finally through F into a counterexample to the base case. The witness

component is iteration and the counterexample component is the same iteration read in reverse, along the sequence of witnesses that iteration produced.

In $\mathrm{Gir}(\mathcal{C})$ the backward map of a morphism does not see the witness, and with it the parameter disappears from the second coordinate. The obstruction can be stated over any base and refuted over sets.

Proposition 4. *A mediating morphism for* $(\mathbb{N}, 0, 0)$ *and the cartesian structure of* $\mathrm{Gir}(\mathcal{C})$ *exists only when* $F = F \circ G$*, and over sets this fails:* $(\mathbb{N}, \emptyset, \emptyset)$ *is not a parametrised natural numbers object for the cartesian structure of* $\mathrm{Gir}(\mathbf{Sets})$*.*

Proof. As above, any candidate is of the form $(\mathbb{N}, 0, 0)$ and $A \times N \cong (U \times \mathbb{N}, X, \alpha')$. A morphism $A \times N \to B$ in $\mathrm{Gir}(\mathcal{C})$ is a pair $h \colon U \times \mathbb{N} \to V$ and $H \colon Y \to X$. The triangle gives $H = F$ and the square gives $H = H \circ G$, so a mediating morphism can exist only when $F = F \circ G$.

Over sets, take $B = (\{a, b\}, \{p, q\}, \beta)$ with $a \,\beta\, p$ and $b \,\beta\, q$ and nothing else, and let (g, G) interchange a with b and p with q. This is a morphism $B \to B$: the two nonvacuous instances of the condition are $a \,\beta\, G(q) = p$, which gives $g(a) = b \,\beta\, q$, and $b \,\beta\, G(p) = q$, which gives $g(b) = a \,\beta\, p$. Now take $A = B$ and $(f, F) = \mathrm{id}$. Then $F = \mathrm{id}$ while $F \circ G = G \neq \mathrm{id}$, and no mediating morphism exists. □

The products of the two categories agree while their natural numbers objects do not. The product of A and B is the same triple $(U \times V, X + Y, \mathrm{ch})$ in $\mathrm{Dial}(\mathbf{Sets})$ and in $\mathrm{Gir}(\mathbf{Sets})$, with the same projections and the same terminal object, and the argument forcing a candidate into the shape $(\mathbb{N}, \emptyset, \emptyset)$ runs identically in both. Without parameters the answers also agree: a morphism from the terminal object to $B = (V, Y, \beta)$ exists only when Y is empty, so every object to be factored through is already trivial in its second coordinate, and $(\mathbb{N}, \emptyset, \emptyset)$ satisfies Lawvere's condition in $\mathrm{Dial}(\mathbf{Sets})$ exactly as it does in $\mathrm{Gir}(\mathbf{Sets})$, which is Proposition 3.1 of [7]. The divergence appears only with the parameter restored, and it is not a divergence in the structure the universal property quantifies over but in the morphisms available to do the factoring. The backward map of a dialectica morphism sees the witness it is refuting and the backward map of a Girard morphism does not, which Proposition 4 turns into the failure of the property. The computations of [7] are made in $\mathrm{Gir}(\mathbf{Sets})$, whose morphisms are the morphisms of relational systems familiar from set theory, and that is the construction which work on cardinal invariants makes the natural one to reach for.

6 Tensors

For the tensors the unit is $I = (1, 1, =)$ rather than $(1, 0, 0)$, and the object that supports zero and successor changes accordingly.

Proposition 5. *The object* $N = (\mathbb{N}, 1, \top)$*, where* $\top$ *is the full relation on* $\mathbb{N} \times 1$*, with* $z = (0, \mathrm{id}_1)$ *and* $s = (s, \mathrm{id}_1)$*, is a parametrised natural numbers object for the tensor of* $\mathrm{Dial}(\mathcal{C})$*.*

Proof. The maps z and s are morphisms: for z the condition reads $* = *$ implies $0 \top *$, and for s it reads $n \top *$ implies $n + 1 \top *$. For $A = (U, X, \alpha)$ we have $A \otimes N = (U \times \mathbb{N},\ X \times 1,\ \alpha \otimes \top) \cong (U \times \mathbb{N}, X, \alpha')$ with $(u, n) \,\alpha'\, x$ iff $u \,\alpha\, x$, which is the object appearing in the proof of Proposition 3, and the two equations unwind to the same recursions for h and H. The verification is then the one given there. □

So in Dial($\mathcal{C}$) the cartesian and the monoidal answers agree, up to the isomorphism $X + 0 \cong X \times 1$ in the counterexample coordinate, and the tensor version needs no coproducts in $\mathcal{C}$ at all. The object $(\mathbb{N}, 0, 0)$ is not available for the tensor: there $A \otimes (\mathbb{N}, 0, 0)$ has initial counterexamples, and a morphism from it to the unit would need a map $U \times \mathbb{N} \times 1 \to 0$. Likewise $(\mathbb{N}, 1, \top)$ admits no zero from the terminal object, since that would need a map $1 \times 1 \to 0$. Each structure has its own object.

Proposition 6. *The object $N = (\mathbb{N}, 1, \top)$, with $z = (0, \mathrm{id}_1)$ and $s = (s, \mathrm{id}_1)$, is a parametrised natural numbers object for $\otimes_{\mathrm{Gir}}$ in* $\mathrm{Gir}(\mathcal{C})$.

Proof. For $A = (U, X, \alpha)$ we have $A \otimes_{\mathrm{Gir}} N = (U \times \mathbb{N},\ X^{\mathbb{N}} \times 1^U,\ \alpha \otimes_{\mathrm{Gir}} \top) \cong (U \times \mathbb{N},\ X^{\mathbb{N}},\ \alpha'')$ with $(u, n)\ \alpha''\ \varphi$ iff $u\ \alpha\ \varphi(n)$, the second conjunct being vacuous. Let $(f, F)\colon A \to B$ and $(g, G)\colon B \to B$, so $F\colon Y \to X$ and $G\colon Y \to Y$. A morphism $A \otimes_{\mathrm{Gir}} N \to B$ is a pair $h\colon U \times \mathbb{N} \to V$ and $H\colon Y \to X^{\mathbb{N}}$ subject to

$$u\ \alpha\ H(y)(n) \implies h(u, n)\ \beta\ y. \tag{2}$$

The two equations give $h(u, 0) = f(u)$ and $h(u, n{+}1) = g(h(u, n))$ in the first coordinate, and

$$H(y)(0) = F(y), \qquad H(y)(n + 1) = H\big(G(y)\big)(n)$$

in the second. Transposing, H is a morphism $\Phi\colon \mathbb{N} \to X^Y$ with $\Phi(0) = F$ and $\Phi(n{+}1) = \Phi(n) \circ G$, which iteration in $\mathcal{C}$ determines, so H is determined, with $H(y)(n) = F(G^n(y))$, and a mediating morphism is unique if it exists.

For (2) we apply Lemma 1 to the subobject of $U \times \mathbb{N}$ given by $\forall y\,(u\ \alpha\ H(y)(n) \Rightarrow h(u, n)\ \beta\ y)$. At 0, if $u\ \alpha\ F(y)$ then $f(u)\ \beta\ y$ and $f(u) = h(u, 0)$. At $n + 1$, suppose $u\ \alpha\ F(G^{n+1}(y)) = F\big(G^n(G(y))\big)$. The induction hypothesis, applied to the counterexample $G(y)$, gives $h(u, n)\ \beta\ G(y)$, and the condition on (g, G) then gives $g(h(u, n))\ \beta\ y$, which is $h(u, n + 1)\ \beta\ y$. □

The counterexample coordinate of the Girard tensor is a sequence. The mediating map sends a counterexample y to the sequence $n \mapsto F(G^n(y))$ of its iterates, and the recursion $H(y)(n + 1) = H(G(y))(n)$ is the equation of a coiteration, with G as the coalgebra structure on Y and F as the map to the carrier. In any cartesian closed category with a natural numbers object, $X^{\mathbb{N}}$ with head and tail is the final coalgebra for $X \times (-)$, uniqueness coming from the transposed iteration in the proof, and H is the unique coalgebra map from $(Y, \langle F, G\rangle)$ into it, which makes precise Charles Morgan's observation that the counterexample maps of zero and successor are the head and the tail of streams. The counterexamples of $A \otimes_{\mathrm{Gir}} N$ are functions $\mathbb{N} \to X$, and this function space, which the cartesian product does not provide, gives the second coordinate somewhere to put the iteration that the backward map of a Girard morphism cannot perform on witnesses.

Each tensor has so far been paired with the morphisms it was built for, and nothing forces that pairing. The Dial tensor is a bifunctor on Gir(**Sets**), since for Girard morphisms (f, F) and (g, G) the pair $(f \times g,\ F \times G)$ satisfies the condition, each component being tested on its own; and $\otimes_{\mathrm{Gir}}$ is a bifunctor on Dial(**Sets**), with backward components $\varphi(v_0) = F(u, \varphi'(g(v_0)))$ and $\psi(u_0) = G(v, \psi'(f(u_0)))$; identities return φ' and ψ' unchanged, and in a composite the witness slot is filled by the same u at both stages, so the functoriality laws hold. Asked across the pairing, the question gives a failure on one side and a weakening on the other.

Proposition 7. *No object of* Gir(**Sets**) *is a parametrised natural numbers object for the Dial tensor.*

Proof. Let $N = (\mathbb{N}, M, \mu)$ carry $z = (0, Z)$ and $s = (+1, S')$, and let $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$. Then $A \otimes N = (U \times \mathbb{N},\ X \times M,\ \alpha \otimes \mu)$, and a morphism $A \otimes N \to B$ in Gir(**Sets**) is a pair $h\colon U \times \mathbb{N} \to V$ and $H = (H_1, H_2)\colon Y \to X \times M$. The backward component of $\mathrm{id}_A \otimes z$ is $\mathrm{id}_X \times Z$ and that of $\mathrm{id}_A \otimes s$ is $\mathrm{id}_X \times S'$, so the triangle gives $H_1 = F$ and the square gives $H_1 = H_1 \circ G$ together with $S' \circ H_2 = H_2 \circ G$. The first two say $F = F \circ G$, which does not mention M, and the object B and the endomorphism (g, G) of Proposition 4 refute it: there $F = \mathrm{id}$ while $F \circ G = G$. □

Proposition 8. *The object* $(\mathbb{N}, 1, \top)$ *is a weak parametrised natural numbers object for* $\otimes_{\mathrm{Gir}}$ *in* Dial(**Sets**)*, the mediating morphism existing but not being unique.*

Proof. For $A = (U, X, \alpha)$ we have $A \otimes_{\mathrm{Gir}} N \cong (U \times \mathbb{N},\ X^{\mathbb{N}},\ \alpha'')$ with (u, n) related to φ iff $u\,\alpha\,\varphi(n)$, as in Proposition 6. A morphism $A \otimes_{\mathrm{Gir}} N \to B$ in Dial(**Sets**) is a pair $h\colon U \times \mathbb{N} \to V$ and $H\colon U \times \mathbb{N} \times Y \to X^{\mathbb{N}}$ subject to $u\,\alpha\,H(u,n,y)(n) \Rightarrow h(u,n)\,\beta\,y$, so only the value of $H(u,n,y)$ at n is tested. The backward component of $\mathrm{id}_A \otimes_{\mathrm{Gir}} z$ sends φ to $\varphi(0)$ and that of $\mathrm{id}_A \otimes_{\mathrm{Gir}} s$ sends φ to $\varphi \circ (+1)$, so besides the recursion for h of Proposition 3 the triangle and the square give

$$H(u,0,y)(0) = F(u,y), \qquad H(u,n+1,y)(k+1) = H\big(u,n,G(h(u,n),y)\big)(k)$$

for all u, n, y and k, and nothing else. Each application of the square lowers both indices by one while transporting the counterexample, so writing $y_0 = y$ and $y_{j+1} = G(h(u, m-j-1), y_j)$, an entry (m,k) is reached from exactly one entry with an index equal to zero, namely $(0, k-m)$ along y_m when $k \geq m$ and $(m-k, 0)$ along y_k when $k < m$. Choose the values $H(u,0,y)(j)$ for $j \geq 1$ and $H(u,m,y)(0)$ for $m \geq 1$ arbitrarily, set $H(u,0,y)(0) = F(u,y)$, and define every other entry by tracing it to the boundary. The square then holds by construction, since $(n+1, k+1)$ and (n,k) trace to the same boundary entry along the same transported counterexamples, and the triangle holds by the choice at the corner. On the diagonal every entry traces to the corner, so

$$H(u,m,y)(m) = F\Big(u,\ G\big(h(u,0),\ G(h(u,1),\ \ldots\ G(h(u,m-1),\ y)\ldots)\big)\Big),$$

which is the backward component of Proposition 3, and the condition holds by the induction given there. Every choice of the boundary values therefore gives a mediating morphism. Taking $A = (1, \{p,q\}, \alpha)$ for any relation α, $B = I$, and for f and g any morphisms $A \to I$ and $I \to I$, which exist since the relation of I is full, two choices of $H(*,1,*)(0)$, one p and one q, give distinct mediating morphisms, so the mediating morphism is not unique. □

A parametrised natural numbers object belongs to a tensor together with a shape of morphism. Collecting the four propositions and the two above,

	Dial(**Sets**)	Gir(**Sets**)
products	$(\mathbb{N}, \emptyset, \emptyset)$	none
Dial tensor	$(\mathbb{N}, 1, \top)$	none
$\otimes_{\mathrm{Gir}}$	$(\mathbb{N}, 1, \top)$ weak	$(\mathbb{N}, 1, \top)$

so each tensor serves the morphisms it was built for and fails or weakens with the other. Two of the three answers in the second column are empty, and the reason is the one the

first column makes visible: a backward map that cannot see the witness has nowhere to put a recursion unless the tensor hands it a function space.

The table is computed over sets, and its three unqualified positive entries hold over any $\mathcal{C}$ as in Section 4, by Propositions 3, 5 and 6.

The three positive entries are copowers. For a set K, the copower of an object A by K is a coproduct of K copies of A, that is, an object $K \cdot A$ with coprojections $\iota_k \colon A \to K \cdot A$ for $k \in K$ such that for every family of morphisms $f_k \colon A \to B$ there is a unique $h \colon K \cdot A \to B$ with $h\iota_k = f_k$ for all k [11]. Burroni's theorem [2] says that a category with countable copowers is a Peano–Lawvere category, the free object over A being the copower $\mathbb{N} \cdot A$ with the first coprojection as zero and the shift as successor, and Charles Morgan observed that this applies to Gir(**Sets**). In Dial(**Sets**) coproducts are in general only weak, but copowers are genuine, since all the copies of A share their counterexamples. In both constructions over sets the countable copowers can be written down, and they are the objects found above.

Proposition 9. *Let $A = (U, X, \alpha)$. In* Dial(**Sets**) *the object $(U \times \mathbb{N},\ X,\ \alpha')$ of Proposition 3 is the countable copower of A, with coprojections $\iota_n = (u \mapsto (u, n),\ (u, x) \mapsto x)$. In* Gir(**Sets**) *the object $(U \times \mathbb{N},\ X^{\mathbb{N}},\ \alpha'')$ of Proposition 6 is the countable copower of A, with coprojections $\iota_n = (u \mapsto (u, n),\ \varphi \mapsto \varphi(n))$.*

Proof. In Dial(**Sets**) the condition on ι_n reads $u\,\alpha\,x \Rightarrow (u, n)\,\alpha'\,x$, which holds by definition of α'. Given morphisms $(f_n, F_n) \colon A \to B$, put $h(u, n) = f_n(u)$ and $H(u, n, y) = F_n(u, y)$. The condition on (h, H) is the conjunction of the conditions on the (f_n, F_n), the composite $(h, H)\iota_n$ is (f_n, F_n), and these equations force h and H, so the copairing exists and is unique. All copies of A share their counterexamples, which is why the coproduct is genuine here, as in the case of coproducts of objects with the same counterexamples in [6]. In Gir(**Sets**) the condition on ι_n reads $u\,\alpha\,\varphi(n) \Rightarrow (u, n)\,\alpha''\,\varphi$, again by definition. Given Girard morphisms (f_n, F_n), put $h(u, n) = f_n(u)$ and $H(y) = (n \mapsto F_n(y))$; the composite with ι_n has backward component $y \mapsto H(y)(n) = F_n(y)$, and the condition on (h, H) is again the conjunction of the conditions on the (f_n, F_n). □

So Burroni's free object over A is $A \times (\mathbb{N}, \emptyset, \emptyset) \cong A \otimes (\mathbb{N}, 1, \top)$ in Dial(**Sets**) and $A \otimes_{\mathrm{Gir}} (\mathbb{N}, 1, \top)$ in Gir(**Sets**), the latter being the object of Charles Morgan's 2013 computation, and the product $A \times (\mathbb{N}, \emptyset, \emptyset)$ of Gir(**Sets**), which is not a copower there, is the one that fails. Over a general $\mathcal{C}$ there need not be countable coproducts, but $U \times \mathbb{N}$ and $X^{\mathbb{N}}$ are objects of $\mathcal{C}$, so the same formulas define objects of Dial($\mathcal{C}$) and Gir($\mathcal{C}$), and Propositions 3, 5 and 6 show that they carry parametrised recursion. We expect them to be copowers in the internal sense, as sums along $\mathbb{N} \to 1$ in the fibration of dialectica objects over $\mathcal{C}$, but that belongs with the fibred treatment of induction and is not proved here.

7 Recursion and corecursion

Since Dial($\mathcal{C}$) has finite products, primitive recursion follows from parametrised iteration by the usual argument, iterating on $B \times N$ and projecting; the witness component of the resulting recursor is the recursor of System T [8].

The pattern hoped for in the earlier note, recursion in one coordinate and corecursion in the other within a single object, is realised by the Girard tensor rather than by the cartesian structure. The counterexample component of Proposition 3 uses the step

morphism once for each stage of the computation, so the recursion it performs is not linear in (g, G). It is also, up to one application, the functional by which Gödel's interpretation handles induction. Jeremy Avigad and Solomon Feferman [1] verify the induction rule by taking witnesses $d(0) = a$ and $d(u+1) = c(u, d(u))$ and transporting a counterexample backwards by $e(y, 0) = y$ and $e(y, z+1) = b(u - z - 1,\, e(y, z))$, where b is the backward component of the step instantiated at the witness reached at that stage; unfolding H gives the same nest, with G for b and $h(u, k)$ for $d(k)$. The one extra application is F, the backward part of the base morphism. When the parameter is the base itself, $A = B$ with (f, F) the identity, F is a projection and H is exactly their e, so this case is the rule as usually stated, and a general parameter gives the rule with an assumption. The terminal object is not the right parameter for this comparison: a morphism from $(1, \emptyset, \emptyset)$ to B exists only when B has no counterexamples, and then the backward recursion has nothing to carry.

Iteration on objects is not yet induction on formulas. An object of Dial(**Sets**) is a predicate, not a family of predicates indexed by the natural numbers, and induction as a rule asks for a section over 0 and a step to yield a section over every n. That belongs in the fibred setting, where predicates live in the fibres of a linear hyperdoctrine over **Sets** and $\mathbb{N}$ is the natural numbers object of the base. The forwards and backwards pair computed here is what one would expect to do the work there.

8 Comonoids

The definition used above is the one Robert Paré and Leopoldo Román gave, and it is obtained from the cartesian definition by replacing the product with the tensor. An object meeting it is never linear in the sense a linear logician would want, whatever the ambient category, and the reason is the universal property itself rather than anything about dialectica objects. Maria Emilia Maietti, Valeria de Paiva and Eike Ritter [5] observed this for lists on the tensor unit in a linear type theory; the statement is categorical and the proof needs only the parameter to be taken trivial.

Proposition 10. *Let $\mathcal{V}$ be a monoidal category with a parametrised natural numbers object N in the sense of Paré and Román, with zero $z\colon I \to N$ and successor $s\colon N \to N$. Then N carries a comonoid structure, with counit $w\colon N \to I$ the unique morphism satisfying $wz = \mathrm{id}_I$ and $ws = w$, and comultiplication $d\colon N \to N \otimes N$ the unique morphism satisfying $dz = (z \otimes z)\lambda_I^{-1}$ and $ds = (s \otimes s)d$. When $\mathcal{V}$ is symmetric the comonoid is cocommutative.*

Proof. Taking the parameter to be I and composing with the unit isomorphism, the universal property reads: for every object A, every $a\colon I \to A$ and every $l\colon A \to A$ there is a unique $h\colon N \to A$ with $hz = a$ and $hs = lh$. Existence of the two structure maps is this property applied twice, to $A = I$ with $a = \mathrm{id}_I$ and $l = \mathrm{id}_I$, and to $A = N \otimes N$ with $a = (z \otimes z)\lambda_I^{-1}$ and $l = s \otimes s$.

For the counit law, both $\rho_N(\mathrm{id}_N \otimes w)d$ and id_N are mediating morphisms for $A = N$, $a = z$ and $l = s$. The identity satisfies the two equations at once. For the composite, $\rho_N(\mathrm{id}_N \otimes w)dz = \rho_N(z \otimes \mathrm{id}_I)\lambda_I^{-1} = z$ since $wz = \mathrm{id}_I$ and $\rho_I = \lambda_I$, while $\rho_N(\mathrm{id}_N \otimes w)ds = \rho_N(s \otimes ws)d = \rho_N(s \otimes w)d = s\,\rho_N(\mathrm{id}_N \otimes w)d$ by $ws = w$ and naturality of ρ. Uniqueness makes the two equal, and the other counit law is the same computation with λ.

For coassociativity, both $(d \otimes \mathrm{id}_N)d$ and $(\mathrm{id}_N \otimes d)d$ are mediating for $A = N \otimes N \otimes N$ with a the threefold zero and $l = s \otimes s \otimes s$, since $(d \otimes \mathrm{id}_N)ds = (ds \otimes s)d = (s \otimes s \otimes$

$s)(d \otimes \mathrm{id}_N)d$ and likewise on the other side. For cocommutativity, the morphisms σd and d are mediating for the same pair, because $\sigma_{I,I}$ is the identity and σ is natural. □

Nothing in the proof looks at the objects, so the conclusion is about the definition and not about any particular model. By Fox's theorem [3] the cocommutative comonoids in a symmetric monoidal category form a cartesian category, with the tensor restricting to the product there, so a Paré–Román natural numbers object is always an object of that cartesian category. Proposition 10 gives this and no more: it does not show that the mediating morphisms are maps of comonoids, which is what would make N a natural numbers object of the cartesian category itself. The notion is linear in the shape of its diagram and not in what the diagram produces.

The tensor objects of Propositions 5 and 6 are retracts of their own !*-versions.* The retractions can be written down directly. In Dial(**Sets**) multisets on a one-element set are the natural numbers, so $!(\mathbb{N}, 1, \top) = (\mathbb{N}, \mathbb{N}, \top)$. The morphism into it is the identity on witnesses with the unique backward map $\mathbb{N} \times \mathbb{N} \to 1$, dereliction is the identity on witnesses with the backward map $\mathbb{N} \times 1 \to \mathbb{N}$ sending $(n, *)$ to the singleton multiset, and the composite is the identity, both conditions holding because the relations are full. In Gir(**Sets**), where the backward map cannot see the witness and the modality takes the function-space form, the !-version is $(\mathbb{N}, \mathbb{N}^{\mathbb{N}}, \top)$, and the same pair works with the constant function at the singleton multiset in place of the singleton. In Dial(**Sets**) the morphism into the !-version is also the one cofreeness of the modality [6] assigns to the comonoid of Proposition 10. Whatever duplication the modality is there to license, these objects already have, so the natural numbers found in this note are not linear. A natural numbers object that is neither duplicable nor discardable needs a different universal property, such as the restricted list objects of [5], and is not treated here.

9 Algebras and coalgebras

Coproducts in Gir(**Sets**) are given by $A+B = (U+V,\ X\times Y,\ \mathrm{ch})$, where $(u,0)\mathrm{ch}(x,y)$ iff $u\alpha x$ and $(v,1)\mathrm{ch}(x,y)$ iff $v\beta y$, with initial object $(\emptyset, 1, \emptyset)$. In Dial(**Sets**) the coproduct is only weak, and is given in [6] by $A+B = (U+V,\ X^U \times Y^V,\ \mathrm{ev})$, where $(u,0)$ is related to (φ,ψ) iff $u\,\alpha\,\varphi(u)$ and $(v,1)$ is related to (φ,ψ) iff $v\,\beta\,\psi(v)$. A copairing exists there, since $H((u,0),z) = (\lambda u'.\, F(u',z),\ \lambda v'.\, G(v',z))$ satisfies both conditions, but the triangles fix only the value of φ at the witness being tested and leave the rest of φ, and all of ψ, free, so it is not unique. When the two objects share their counterexamples, however, the coproduct is genuine and is simply $A + B = (U + V,\ X,\ \mathrm{ch})$ with $(u,0)\,\mathrm{ch}\,x$ iff $u\,\alpha\,x$ and $(v,1)\,\mathrm{ch}\,x$ iff $v\,\beta\,x$, the copairing being $H((u,0),z) = F(u,z)$ and $H((v,1),z) = G(v,z)$, which the triangles force. The functors below are given by these formulas rather than by a universal property on the Dial side, which is all an initial algebra needs, and the particular coproduct appearing in Proposition 12 is of the genuine kind, since there I and $(\mathbb{N}, 1, \top)$ both have 1 as their counterexamples.

Proposition 11. *The object* $(\mathbb{N}, \emptyset, \emptyset)$ *is the initial algebra for* $T + (-)$ *in* Dial(**Sets**) *and in* Gir(**Sets**)*, where* $T = (1, \emptyset, \emptyset)$ *is the terminal object.*

Proof. Computing, $T + A = (1 + U,\ \emptyset \times X,\ \emptyset)$ in Gir(**Sets**) and $(1 + U,\ \emptyset^1 \times X^U,\ \mathrm{ev})$ in Dial(**Sets**), which is $(1 + U, \emptyset, \emptyset)$ either way. An algebra structure $(b, B)\colon T + A \to A$ then has a backward component landing in the empty set, from X in Gir(**Sets**) and from $(1 + U) \times X$ in Dial(**Sets**), so X is empty. Conversely every object $(U, \emptyset, \emptyset)$ with a

point and an endomap of U carries one, and a morphism between two such algebras has a backward component out of the empty set, so the category of algebras is isomorphic to the category of $1 + (-)$-algebras in **Sets**, whose initial object is $\mathbb{N}$ with zero and successor. □

Proposition 12. *The object* $(\mathbb{N}, 1, \top)$ *is the initial algebra for* $I + (-)$ *in* Dial(**Sets**) *and in* Gir(**Sets**)*, where* $I = (1, 1, =)$ *is the unit of the tensor.*

Proof. Take Gir(**Sets**) first. Computing, $I + A = (1 + U,\ 1 \times X,\ \mathrm{ch}) \cong (1 + U,\ X,\ \mathrm{ch}')$ with $(*, 0)\ \mathrm{ch}'\ x$ for every x, and $(u, 1)\ \mathrm{ch}'\ x$ iff $u\ \alpha\ x$. An algebra structure (b, B) has $b\colon 1 + U \to U$ and $B\colon X \to X$, and its condition splits in two. At $(*, 0)$ the hypothesis holds for every x, so the point $u_0 = b(*, 0)$ satisfies $u_0\ \alpha\ x$ for all x in X, which is Fact 1 of [7] arriving as the algebra condition. At $(u, 1)$ it reads $u\ \alpha\ B(x) \Rightarrow g(u)\ \alpha\ x$, writing $g(u) = b(u, 1)$.

The object $(\mathbb{N}, 1, \top)$ carries such a structure, with $b = [0, +1]$ and $B = \mathrm{id}_1$: the first condition asks $0 \top *$ and the second asks that $n \top *$ give $n + 1 \top *$. Let A be any algebra. A morphism $(h, H)\colon (\mathbb{N}, 1, \top) \to A$ has $H\colon X \to 1$ unique, and the algebra square forces $h(0) = u_0$ and $h(n + 1) = g(h(n))$, so a mediating morphism is unique if it exists. Its condition has a hypothesis that always holds, so what must be checked is that $h(n)\ \alpha\ x$ for every n and x. At 0 this is $u_0\ \alpha\ x$. At $n + 1$ the induction hypothesis gives $h(n)\ \alpha\ B(x)$, and the second algebra condition then gives $g(h(n))\ \alpha\ x$.

In Dial(**Sets**) the computation is $I + A = (1 + U,\ 1^1 \times X^U,\ \mathrm{ev}) \cong (1 + U,\ X^U,\ \mathrm{ev})$, where $(*, 0)$ is related to every φ and $(u, 1)$ is related to φ iff $u\ \alpha\ \varphi(u)$. An algebra structure has $B\colon (1 + U) \times X \to X^U$, its condition at $(*, 0)$ again forces $u_0\ \alpha\ x$ for every x, and at $(u, 1)$ it reads $u\ \alpha\ B((u, 1), x)(u) \Rightarrow g(u)\ \alpha\ x$. Since $1^{\mathbb{N}} \cong 1$, the object $(\mathbb{N}, 1, \top)$ carries the corresponding structure, the mediating morphism is forced as above, and the induction on n, whose statement quantifies over all counterexamples, applies the hypothesis at $B((h(n), 1), x)(h(n))$. □

Negation is available in Gir(**Sets**) *and not in* Dial(**Sets**). Writing $A^\perp = (X, U, \alpha^\perp)$ with $x\ \alpha^\perp\ u$ iff not $u\ \alpha\ x$, the assignment $A \mapsto A^\perp$ is the contravariant involution of [6], under which a morphism $A \to B$ contraposes to one $B^\perp \to A^\perp$ with the two components exchanged. It is available there because the backward map of a Girard morphism carries no witness; in Dial(**Sets**) it carries one, and turning the pair around would make the forward map of the contraposed morphism depend on a counterexample. The clauses needed below are read off the definitions: $(A \times B)^\perp$ and $A^\perp + B^\perp$ are the same triple, $T^\perp$ is the initial object, and $I^\perp = J$ because equality on a one-element set has empty complement.

Involutivity is the only place in this note where the ambient is used classically, since $u\ \alpha^{\perp\perp}\ x$ unfolds to the double negation of $u\ \alpha\ x$ and the relations are two-valued, so $\alpha^{\perp\perp}$ and α are the same subset of $U \times X$. Over a Heyting algebra or over a lineale whose negation is not involutive one has $\alpha \leq \alpha^{\perp\perp}$ instead, and the assignment gives a map $A \to A^{\perp\perp}$, so the duality below becomes a comparison in one direction; the strong negations of the dialectica categories over a Heyting algebra are of that kind.

Proposition 13. *The object* $(1, \mathbb{N}, \emptyset)$ *is the final coalgebra for* $J \times (-)$ *in* Dial(**Sets**) *and in* Gir(**Sets**)*, where* $J = (1, 1, \emptyset)$.

Proof. Computing, $J \times A = (1 \times U,\ 1 + X,\ \mathrm{ch}) \cong (U,\ 1 + X,\ \mathrm{ch})$ with $u\ \mathrm{ch}\ (*, 0)$ for no u, since the relation of J is empty, and $u\ \mathrm{ch}\ (x, 1)$ iff $u\ \alpha\ x$. A coalgebra structure

$(k, K)\colon A \to J \times A$ in Dial(**Sets**) has $k\colon U \to U$ and $K\colon U \times (1 + X) \to X$, and its condition splits in two: at $(*, 0)$ the conclusion is false, so u must fail $K(u, (*, 0))$, and at $(x, 1)$ it reads $u \,\alpha\, K(u, (x, 1)) \Rightarrow k(u) \,\alpha\, x$.

The object $(1, \mathbb{N}, \emptyset)$ carries such a structure, with k the identity and $K(*, (*, 0)) = 0$, $K(*, (n, 1)) = n+1$, both conditions holding because its relation is empty. Let A be any coalgebra. A morphism $(h, H)\colon A \to (1, \mathbb{N}, \emptyset)$ has $h\colon U \to 1$ unique, and the coalgebra square forces

$$H(u, 0) = K(u, (*, 0)), \qquad H(u, n + 1) = K\big(u, (H(k(u), n), 1)\big),$$

so a mediating morphism is unique if it exists. Since the relation of $(1, \mathbb{N}, \emptyset)$ is empty, its condition asks that u fail $H(u, n)$ for every u and n. At 0 this is the first coalgebra condition. The induction on n has a statement quantifying over all witnesses, so at $n + 1$ the hypothesis may be taken at $k(u)$, where it says that $k(u)$ fails $H(k(u), n)$; the second coalgebra condition then gives that u fails $K(u, (H(k(u), n), 1))$, since otherwise $k(u)$ would be related to $H(k(u), n)$.

In Gir(**Sets**) the backward components lose their witness argument and the same computation applies. The statement there is also the image of Proposition 12 under the negation recalled above, since $(\mathbb{N}, 1, \top)^{\perp} = (1, \mathbb{N}, \emptyset)$ and $I^{\perp} = J$. □

Initiality does not imply parametrised recursion here. The object $(\mathbb{N}, \emptyset, \emptyset)$ is the initial algebra for $T + (-)$ in Gir(**Sets**) by Proposition 11 and is not a parametrised natural numbers object there by Proposition 4, so the two notions come apart. The same object is both in Dial(**Sets**), by Propositions 3 and 11, so the separation is not an artefact of the functor but a difference between the two constructions, and a reader who takes an initial algebra to deliver recursion with parameters, as it does in a topos, should not take it to do so here.

10 Conclusions

The Dialectica interpretation runs on System T, and Dial($\mathcal{C}$) carries the recursion that goes with it, for any cartesian closed category $\mathcal{C}$ with a natural numbers object whose subobjects carry intuitionistic first-order logic. The object $(\mathbb{N}, 0, 0)$ is a parametrised natural numbers object for its products and $(\mathbb{N}, 1, \top)$ is one for its tensor, and in both the witness coordinate iterates while the counterexample coordinate recurs backwards along the witnesses that iteration produced, which is available because the backward map of a dialectica morphism sees the witness it is refuting. In Gir($\mathcal{C}$) the cross tensor has such an object, its counterexample coordinate being the final coalgebra of streams, while over sets the products have none, so the same question answers differently in the two constructions of [6], and the difference lies in the shape of the backward map rather than in the structure the universal property quantifies over. Over sets all three objects are countable copowers, as Burroni's theorem leads one to expect, and over $\mathcal{C}$ they are given by the same formulas without any infinite coproducts.

The monoidal definition carries its own limit. Any object satisfying it in a symmetric monoidal category is a cocommutative comonoid, so $(\mathbb{N}, 1, \top)$ is duplicable and discardable in both constructions and is a retract of its own !-version, as Maria Emilia Maietti, Valeria de Paiva and Eike Ritter [5] observed for lists on the tensor unit. The recursion the dialectica categories carry is Gödel's, and the numbers it runs on are not linear ones.

Algebras and coalgebras are available in both categories. The objects $(\mathbb{N}, \emptyset, \emptyset)$ and $(\mathbb{N}, 1, \top)$ are the initial algebras for $T+(-)$ and $I+(-)$, and $(1, \mathbb{N}, \emptyset)$ is the final coalgebra

for $J \times (-)$, the last of these being the de Morgan dual of the second in Gir(**Sets**), where negation is an involution. Initiality does not deliver recursion with parameters here, as $(\mathbb{N}, \emptyset, \emptyset)$ shows in Gir(**Sets**), and the parametrised property is the one the interpretation asks for.

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